

RANONG, Thailand

WMO: 485320

Lat: 9.778N

Lon: 98.585E

Elev: 57

StdP: 14.67

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	70.4	71.8	62.1	83.6	79.0	64.1	89.9	79.9	13.7	83.0	11.7	84.2	0.9	60	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	15.6	95.6	78.1	94.3	78.1	93.0	78.0	81.3	89.2	80.8	88.7	80.2	87.9	5.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	151.8	84.9	78.8	149.9	84.6	78.3	147.1	84.1	45.1	89.4	44.5	89.3	43.9	88.5	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.9	10.6	9.1	66.1	98.2	2.3	2.2	64.5	99.8	63.1	101.1	61.8	102.3	60.1	103.9		
			63.1	83.3	2.5	1.6	61.3	84.4	59.9	85.3	58.5	86.2	56.7	87.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.0	81.1	82.4	84.0	84.8	83.3	82.0	81.4	81.2	80.8	81.0	81.3	80.7
	DBStd	2.43	1.83	1.64	2.04	1.99	2.54	2.25	2.18	2.20	2.08	1.88	1.86	1.91
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11684	964	908	1055	1045	1032	961	974	968	924	962	939	952
	CDD65	6209	499	488	590	595	567	511	509	503	474	497	489	487
	CDH74	63265	4935	5374	6957	7080	6093	5218	4952	4835	4265	4438	4592	4526
	CDH80	23708	1989	2511	3399	3315	2294	1682	1462	1386	1174	1348	1568	1580
Wind	WSAvg	2.4	2.6	2.5	2.4	2.2	2.2	2.6	2.7	2.9	2.2	1.7	1.9	2.9
Precipitation	PrecAvg	165.4	0.9	0.6	2.9	5.6	20.1	26.1	27.9	28.9	26.9	17.2	5.9	2.4
	PrecMax	211.7	9.7	3.0	16.5	27.0	39.1	40.0	53.4	50.0	45.3	43.1	18.0	12.6
	PrecMin	126.6	0.0	0.0	0.0	0.0	8.2	12.4	14.3	10.9	12.0	4.6	1.1	0.0
	PrecStd	24.3	1.9	0.8	3.9	5.4	8.3	7.9	8.7	9.4	8.7	8.6	4.1	2.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.9	95.3	97.7	97.7	96.0	92.0	90.7	90.0	90.6	91.4	92.0	91.7
		MCWB	75.6	76.1	77.2	78.3	79.8	79.3	78.8	78.6	79.0	78.4	78.1	77.1
	2%	DB	91.5	93.7	95.7	95.6	93.5	90.5	89.4	89.1	89.2	89.8	90.7	90.4
		MCWB	75.7	76.3	77.3	79.1	80.1	79.2	78.9	78.6	78.7	78.5	78.0	76.6
	5%	DB	90.1	92.4	94.2	94.0	91.4	89.3	88.1	88.0	87.8	88.5	89.5	89.1
		MCWB	75.7	76.2	77.3	79.3	79.8	79.1	78.8	78.6	78.5	78.4	77.8	76.3
	10%	DB	88.9	91.1	92.7	92.5	89.6	87.8	86.8	86.7	86.2	86.9	87.9	87.6
		MCWB	75.6	76.0	77.4	79.2	79.5	78.9	78.6	78.6	78.2	78.2	77.6	76.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.6	79.9	80.8	82.1	82.2	81.5	81.0	80.9	80.6	81.0	80.5	80.0
		MCDB	88.2	90.2	91.6	91.5	90.6	87.8	86.4	86.1	87.1	87.4	88.2	87.5
	2%	WB	78.2	78.7	79.8	81.0	81.2	80.6	80.1	80.0	79.7	79.9	79.6	78.8
		MCDB	87.3	89.6	90.7	90.9	89.4	87.1	86.0	85.6	86.1	86.6	87.4	87.0
	5%	WB	77.2	77.9	79.1	80.5	80.5	79.9	79.5	79.2	79.1	79.2	78.9	77.9
		MCDB	86.7	88.9	90.0	90.4	88.6	86.3	85.5	85.2	85.5	85.9	86.8	86.0
	10%	WB	76.4	77.0	78.4	79.9	79.9	79.2	79.0	78.8	78.5	78.6	78.2	77.0
		MCDB	85.8	88.1	89.1	89.5	87.3	85.5	85.0	85.0	84.6	85.1	85.6	85.0
Mean Daily Temperature Range	5% DB	MDBR	16.3	18.2	17.8	15.6	11.9	10.3	9.8	9.6	10.1	11.4	12.6	13.9
		MCDBR	18.3	20.1	20.3	17.9	14.8	12.8	12.0	11.9	12.7	14.4	15.8	16.5
		MCWBR	6.4	6.6	6.4	5.6	5.0	4.5	4.4	4.4	4.7	5.7	5.9	6.2
	5% WB	MCDBR	16.4	18.2	18.0	16.3	13.2	11.3	10.5	10.5	11.4	12.9	14.0	14.6
		MCWBR	6.6	7.1	6.7	5.9	5.1	4.9	4.7	4.8	4.9	5.8	5.9	6.2
Clear-Sky Solar Irradiance	taub	0.482	0.498	0.506	0.480	0.457	0.459	0.460	0.451	0.434	0.453	0.475	0.481	
	taud	2.200	2.146	2.169	2.249	2.358	2.379	2.360	2.380	2.422	2.349	2.263	2.215	
	Ebn at Noon	263	263	262	266	267	263	264	270	278	272	263	260	
	Edh at Noon	46	50	49	45	40	38	39	39	38	40	43	45	
All-Sky Solar Radiation	RadAvg	1584	1846	1895	1868	1495	1402	1352	1390	1409	1375	1389	1390	
	RadStd	136	109	160	119	136	97	126	137	137	144	124	123	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.31	+0.76	N/A	N/A	N/A	+0.36	N/A	N/A	N/A	N/A

Nomenclature: See separate page